Factoring $x^2 + 5x + 6$ into $(x + 2)(x + 3)$ with Prime Factor Blocks

Polynomial Equations & Factoring • Standard: CCSS.MATH.HSA.APR.A.1

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1. MATHEMATICAL CONCEPT & STACK APPROACH: Factor the constant term with Prime Factor blocks, sketch the $(x + p)(x + q)$ area grid, and verify with the SC-32 [POLY] solver.

A polynomial $ax^2 + bx + c$ mirrors base-10 place value with variable powers (x^2, x^1, x^0) in place of 10^2, 10^1, 10^0. Factoring $x^2 + 5x + 6$ into $(x + 2)(x + 3)$ decomposes the constant term into prime factors ($2 \times 3 = 6$) whose sum equals the linear coefficient ($2 + 3 = 5$), which [POLY] verifies when [a, b, c] are pushed into Z, Y, X.

2. MODELING WITH BLOCKS (WHEN APPLICABLE) & STUDENT PREDICTION

Published Block Kit: Prime Factor Studio (PRIME-01) & Algebraic Area Diagram

Hands-On Setup: Stack Prime Factor blocks [2] and [3] ($2 \times 3 = 6$) in the comparator lid and sketch the 2×2 area grid for $(x + 2)(x + 3) = x^2 + 3x + 2x + 6$.

Student Prediction (Before Calculator): Prime factors of 6 are 2 and 3, and their sum is $2 + 3 = 5$. The factored form is $(x + 2)(x + 3)$ with roots -2 and -3.

3. CALCULATOR EXECUTION (SC-32 KEYCAPS & 4-LEVEL STACK TRACE)

Keystroke Flow: `1` `ENTER` `5` `ENTER` `6` `↗` `POLY`

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
`1`	1	Coefficient a = 1 entered into X
`ENTER`	1	1 lifted to Y
`5`	5	Coefficient b = 5 entered into X
`ENTER`	5	b in Y, a in Z
`6`	6	Coefficient c = 6 entered into X
`↗` `POLY`	-2	Press [g:] [POLY]: Root 1 = -2 in Y (factor: x + 2), Root 2 = -3 in X (factor: x + 3)!

4. STORY CONTEXT (From Al-Khwarizmi's Squares to the 1535 Duel in Milan & Kahan's [POLY] Solver)

In 820 AD in Baghdad, Al-Khwarizmi solved $x^2 + 10x = 39$ by completing the square—and in 1535 in Milan, Niccolò Tartaglia defeated Antonio Fiore in a 30-problem public algebra duel overnight! By mapping polynomial coefficients onto the Place Value Tray and factoring constant terms with Prime Factor Studio blocks, students discover how $(x + a)(x + b)$ connects directly to the SC-32 [POLY] solver.

Core Investigation (Student Warm-Up A):

Place coefficients [1], [7], [12] for $x^2 + 7x + 12$ on your tray, factor 12 with Prime Factor blocks to find the pair summing to 7, then verify with [POLY].

Signed Factor Pairs: Factoring Trinomials with Negative Terms

Factoring Quadratic Trinomials • Standard: CCSS.MATH.HSA.REI.B.4

1. EXTENSION CONCEPT (WARM-UP B), BLOCKS & PREDICTION

Concept & Stack Approach: When a quadratic has a negative constant or linear term (such as $x^2 - x - 6$), the factor pair must have opposite signs (-3 and +2) so their product is -6 and their sum is -1. Entering signed coefficients with [+/-] onto the stack verifies both real roots.

Blocks & Prediction: Set up coefficients for $x^2 - x - 6$ on the Place Value Tray and factor $|c| = 6$ into Prime Factor blocks [2] and [3]. Prediction: Assign signs to the factor pair (2, -3) so their product is -6 and their sum is $2 + (-3) = -1$: factors are $(x - 3)(x + 2)$.

2. EXTENSION CALCULATOR SEQUENCE & STACK TRACE

[1] [ENTER] [1] [+/-] [ENTER] [6] [+/-] [↷] [POLY]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
[1]	1	Coefficient a = 1
[ENTER]	1	1 lifted to Y
[1]	1	Enter magnitude of b
[+/-]	-1	Coefficient b = -1 in X
[ENTER]	-1	b in Y, a in Z
[6]	6	Enter magnitude of c
[+/-]	-6	Coefficient c = -6 in X
[↷] [POLY]	3	Press [g:] [POLY]: Root 1 = 3 in Y (factor: x - 3), Root 2 = -2 in X (factor: x + 2)!

3. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Expanding $(x + 2)(x + 3)$ as $x^2 + 6$ by forgetting the two linear cross-terms (2x and 3x).

Hands-On Block Remediation: Place coefficients [1], [5], [6] in the x^2, x^1, x^0 sockets of the Place Value Tray and factor 6 into Prime Factor blocks [2] and [3] so students see that $b = 2 + 3 = 5$ and $c = 2 \times 3 = 6$.

Misconception 2: Using the subtraction operator [−] instead of the change-sign key [+/−] when entering negative polynomial coefficients into [POLY].

Hands-On Block Remediation: Point out that [−] immediately subtracts X from Y, whereas [+/−] negates the number currently in Register X without collapsing the stack.

4. EXTENSION MATHEMATICAL INSIGHT

When a quadratic has a negative constant or linear term (such as $x^2 - x - 6$), the factor pair must have opposite signs (-3 and +2) so their product is -6 and their sum is -1. Entering signed coefficients with [+/-] onto the stack verifies both real roots.

Algebraic Expressions, Base-x Polynomials & Quadratic Factoring

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: QUADRATIC AREA MODEL & PRIME FACTORING RULES

- A polynomial $ax^2 + bx + c$ mirrors base-10 place value with powers (x^2, x^1, x^0) in place of 10^2, 10^1, 10^0.
- To factor $x^2 + bx + c$ into $(x + p)(x + q)$, use Prime Factor Studio blocks to factor c into pairs (p, q) whose sum equals b.
- Signed coefficients: track negative linear or constant terms when testing factor pairs ($p + q = b$ and $p \times q = c$).

Key Sequence: `1` `ENTER` `5` `ENTER` `6` `↷` `POLY`

Investigation 1: Factor $x^2 + 6x + 8$ by factoring 8 with Prime Factor blocks, then verify with [POLY].

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Stack Plan:

Algebraic Setup:

Stack Registers (Y, X):

RPN Keystroke Plan:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Investigation 2: Factor $x^2 - 4$ (difference of two squares) where $b = 0$.

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Stack Plan:

Algebraic Setup:

Stack Registers (Y, X):

RPN Keystroke Plan:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Investigation 3: Factor $x^2 + 5x + 4$ using Prime Factor blocks and verify.

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Stack Plan:

Algebraic Setup:

Stack Registers (Y, X):

RPN Keystroke Plan:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Investigation 4: Factor $x^2 - 5x + 6$ using signed factor pairs.

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Stack Plan:

Algebraic Setup:

Stack Registers (Y, X):

RPN Keystroke Plan:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Unit 08: Algebraic Expressions, Base-x Polynomials & Quadratic Factoring — Answer Key

Complete solutions and SC-32 verification keystrokes for Grades 7–8

1. CORE (WARM-UP A) & EXTENSION (WARM-UP B) SOLUTIONS

Core Investigation (Warm-Up A): Place coefficients [1], [7], [12] for $x^2 + 7x + 12$ on your tray, factor 12 with Prime Factor blocks to find the pair summing to 7, then verify with [POLY].

Core Solution: $(x + 3)(x + 4)$, roots -3 and -4. Keystrokes: [1] [ENTER] [7] [ENTER] [1][2] [g:][POLY]

Core Check Keys: [1] [ENTER] [7] [ENTER] [1] [2] [↗] [POLY]

Extension Challenge (Warm-Up B): Factor $x^2 - 5x + 6$ using Prime Factor blocks for $c = 6$ and verify on the calculator.

Extension Solution: $(x - 2)(x - 3)$, roots 2 and 3. Keystrokes: [1] [ENTER] [5] [+/-] [ENTER] [6] [g:][POLY]

Ext. Check Keys: [1] [ENTER] [5] [+/-] [ENTER] [6] [↗] [POLY]

2. STUDENT LAB WORKSHEET SOLUTIONS (INVESTIGATIONS 1–4)

Investigation 1: Factor $x^2 + 6x + 8$ by factoring 8 with Prime Factor blocks, then verify with [POLY].

Expected Answer: $(x + 2)(x + 4)$; roots -2, -4

SC-32 Verification Keys: [1] [ENTER] [6] [ENTER] [8] [↗] [POLY]

Investigation 2: Factor $x^2 - 4$ (difference of two squares) where $b = 0$.

Expected Answer: $(x - 2)(x + 2)$; roots 2, -2

SC-32 Verification Keys: [1] [ENTER] [0] [ENTER] [4] [+/-] [↗] [POLY]

Investigation 3: Factor $x^2 + 5x + 4$ using Prime Factor blocks and verify.

Expected Answer: $(x + 1)(x + 4)$; roots -1, -4

SC-32 Verification Keys: [1] [ENTER] [5] [ENTER] [4] [↗] [POLY]

Investigation 4: Factor $x^2 - 5x + 6$ using signed factor pairs.

Expected Answer: $(x - 2)(x - 3)$; roots 2, 3

SC-32 Verification Keys: [1] [ENTER] [5] [+/-] [ENTER] [6] [↗] [POLY]

Hands-On Block & Stack Verification Protocol

Students first build the physical block model and write their mathematical strategy and prediction before verifying their result on the StackCalc calculator.